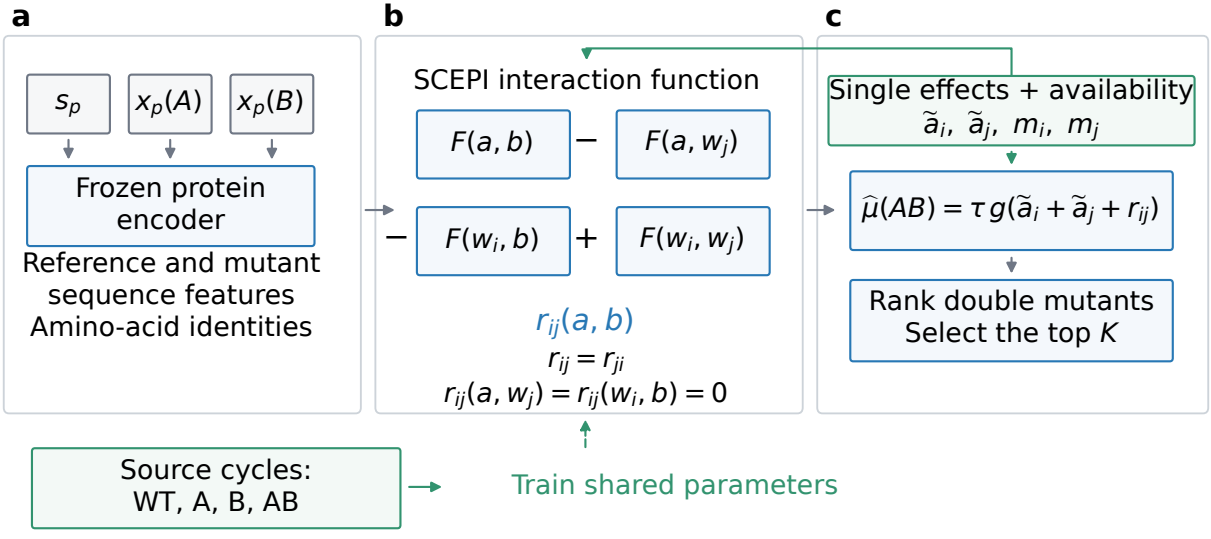
a
s_p
$x_p(A)$
$x_p(B)$
Frozen protein encoder
Reference and mutant sequence features
Amino-acid identities
b
SCEPI interaction function
$F(a,b)$
$-$
$F(a,w_j)$
$-$
$F(w_i,b)$
$+$
$F(w_i,w_j)$
$r_{ij}(a,b)$
$r_{ij}=r_{ji}$
$r_{ij}(a,w_j)=r_{ij}(w_i,b)=0$
c
Single effects + availability
$\tilde{a}_i,\ \tilde{a}_j,\ m_i,\ m_j$
$\hat{\mu}(AB)=\tau\, g(\tilde{a}_i+\tilde{a}_j+r_{ij})$
Rank double mutants
Select the top K
Source cycles:
WT, A, B, AB
Train shared parameters